

Long-term beta blockers for stable angina: systematic review and meta-analysis

De Fen Shu¹, Bi Rong Dong¹, Xiu Fang Lin¹, Tai Xiang Wu² and Guan Jian Liu²

Abstract

Objectives: To assess the effects of long-term beta blockers in patients with stable angina.

Methods: We reviewed the literature up to June 2010 from CENTRAL, MEDLINE, EMBASE, CBM, and CNKI for randomized controlled trials. The appropriate data were meta-analysed using Revman 5.0.

Results: Twenty-six trials including 6108 patients were identified. The treatment with beta blockers has significantly decreased all-cause mortality when compared with no control (OR 0.40, 95% CI 0.20 to 0.79), but has had no statistically differences when compared with placebo (OR 0.92, 95% CI 0.62 to 1.38) and with calcium-channel blocker (CCB) (OR 0.84, 95% CI 0.49 to 1.44). This was similar in patients with fatal and non-fatal acute myocardial infarction when compared with placebo (OR 0.82, 95% CI 0.57 to 1.17) or CCB (OR 1.08, 95% CI 0.71 to 1.66); on revascularization and quality of life. The beta blockers reduced the incident of unstable angina compared to no treatment (OR 0.14, 95% CI 0.07 to 0.29), but increased unstable angina compared to placebo (OR 3.32, 95% CI 1.50 to 7.36). There was a significant reduction of nitrate consumption when beta blockers were compared with CCBs (OR -1.18, 95% CI -1.54 to -0.82), but not with placebo and trimetazidine. There was no significant difference in angina attack between each group. Side effects in beta blocker were similar with ones in controls.

Conclusions: Beta blockers may decrease the death and unstable angina when compared with no treatment, but no more effective than other anti-anginal agents on prophylaxis of myocardial ischaemia in stable angina patients.

Keywords

Beta blocker, stable angina, meta-analysis

Received 6 January 2011; accepted 12 April 2011

Introduction

Ischaemic heart disease is one of the most common causes of death in the Western countries, where it amounts for almost 33% of overall mortality.¹ Crude incidence rate of acute coronary syndrome was 234 per 100,000 person-years.² The economic burden has also increased in each country. The USA spent an estimated \$368 billion in relation to direct and indirect costs of cardiovascular disease (CVD) in 2004.³

Angina pectoris is the result of myocardial ischaemia (MI) caused by an imbalance between myocardial blood supply and oxygen demand. The main goals of treatment in CVD are relief of angina pectoris, slowing the progression of disease, and reduction of future events, especially MI and death.⁴ Antianginal medications include beta blockers, calcium channel blockers, and nitrates.

Beta blockers are used for symptomatic relief of angina and prevention of ischaemic events. These agents inhibit the activation of beta receptors, slow atrioventricular conduction, and decrease myocardial oxygen demand. It has been shown that beta blockers reduce the mortality and morbidity following acute MI (AMI).⁴ The recent guides suggest that beta blockers are the best agents in the treatment of MI because they reduce post-MI mortality.^{5,6} Despite the increasing

¹Geriatrics, West China Hospital, Sichuan University, Chengdu, China.

²Chinese Cochrane Centre, Chinese EBM Centre, West China Hospital, Sichuan University, Chengdu, China.

Corresponding author:

Bi Rong Dong, Geriatrics, West China Hospital, Sichuan University, Chengdu, China

Email: birongdong@163.com

emphasis on secondary prevention of CVD, it is a remaining concern that most CVD patients are still not achieving the recommended goals.⁷

The effects on prognosis in patients with stable angina have not been specifically studied in large trials. Recently, a lot of trials of beta blockers for secondary prevention in ischaemic heart disease have been undertaken.^{8–10} A meta-analysis¹¹ comparing beta blockers, calcium-channel blockers (CCBs), and nitrates for stable angina pectoris found no significant differences in cardiac deaths and AMI, but this finding is based on information from an article published 13 years ago. We have hypothesized that beta blockers are more effective for anti-anginal treatment. The aim of this systematic review study is to assess effects of beta blockers in the treatment of patients with stable angina pectoris by reviewing and analysing literatures.

Methods

Inclusion and exclusion criteria

Types of studies. RCTs with parallel design and an observational period equal to or longer than 4 weeks were included in this study.

Types of participants. Participants in this review were adult patients with stable chest pain syndrome and diagnosed or suspected with ischaemic heart disease.

Types of interventions. Beta blockers were compared with other anti-angina pectoris drugs, placebo, or no intervention. The different doses and administrative routes of beta blockers have been investigated.

Types of outcome. Primary outcomes were all-cause mortality and/or fatal and non-fatal AMI. Secondary outcomes were: (1) resolution of angina symptoms; (2) frequency of angina symptoms; (3) use of nitroglycerine; (4) quality of life (QOL); (5) development of unstable angina (UA) or hospitalization for UA; (6) requirement for additional pharmacological therapy; and (7) hospitalization for revascularization. Any adverse events of the treatment have been studied.

Literature search and screening

A comprehensive and exhaustive search strategy was formulated. The literatures used in this study were obtained from the CENTRAL, MEDLINE, and EMBASE. The Chinese literatures were received from the Chinese Biomedical Literature Database (CBM), Chinese National Knowledge Infrastructure (CNKI) and WANFANG database. The literatures have been

used in this study ranged from the database establishment to June 2010.

The documents were screened by two individual investigators independently by reviewing the title, abstract, and full text. If any data was missed, the original authors will be generally sought.

Quality assessment of included studies

The risk of bias of each trial was assessed and classified it as 'low risk', 'moderate risk', or 'high risk' according to the guidelines of the *Cochrane Handbook for Systematic Reviews of Interventions* 5.0.0.¹²

Data extraction and analysis

The data were extracted by two reviewers independently according to special data extract form. The original authors were contacted in order to find missing data.

Data were combined using RevMan 5.0 and meta-analysed if they are available of sufficient quality and sufficiently similar by using the random-effects model. Odds ratio (OR) or weighted mean differences (WMD) were used for binary or continuous outcomes respectively with 95% confidence interval (CI). Heterogeneity was analysed with an alpha of 0.05 used and with the I^2 test. Possible sources of heterogeneity were assessed by sensitivity and subgroup analyses. A funnel plot of all included trials will be used to assess the presence of publication bias.¹²

Results

Description of studies

Results of the search. The 5512 potentially relevant 'English' articles and 2214 'Chinese' articles were found, with three of them both in English and Chinese. After screening, 26 trials were enrolled in this study. The details of the selection process are exhibited in Figure 1.

Characteristic of included studies. The trials included in this study were published between 1979 and 2008. Details of the characteristics and methodological quality of the included studies are shown in Table 1.

Populations. In total, 6108 patients were enrolled and 5340 of them completed in the investigation. The number of patients enrolled in each trial ranged from 19¹³ to 1460,¹⁴ with nine studies less than 100 patients. Only in three trials were there more than 500 randomized patients.^{8,9,14} The mean age in most trials was older than 50 years old (range 18–83). Twenty-one per cent

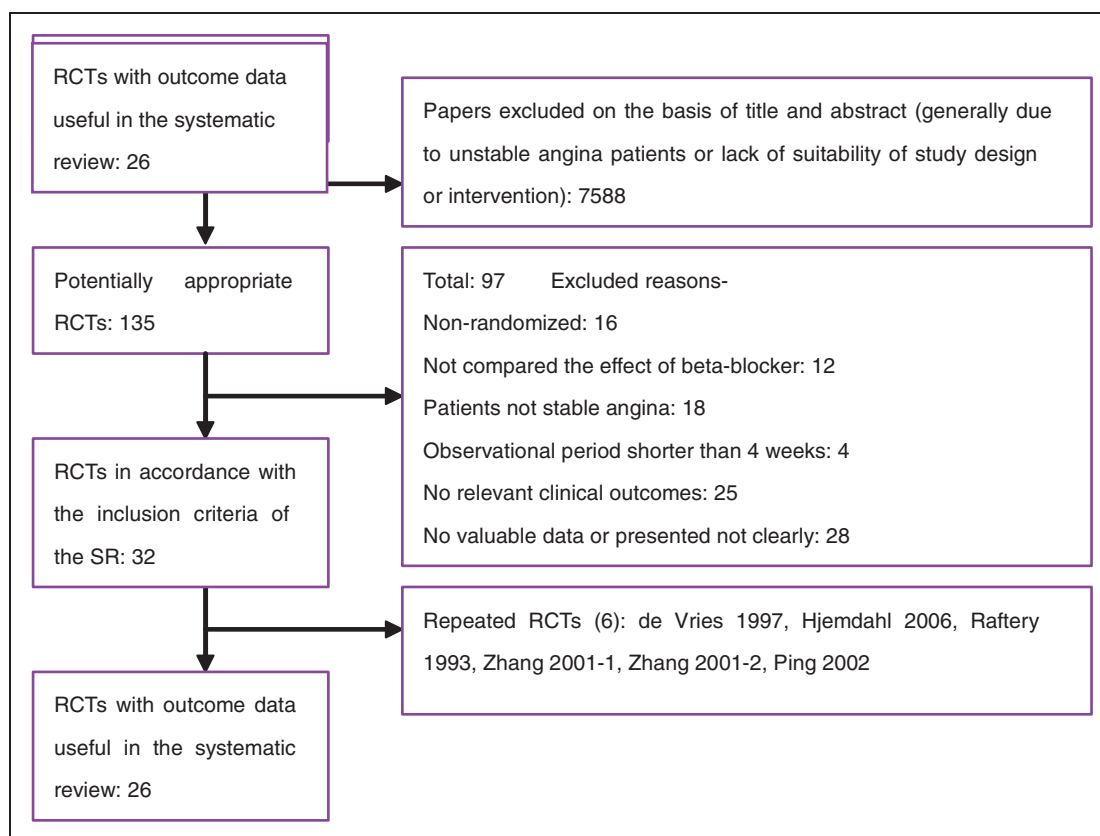


Figure 1. Flow diagram of systematic review.

of the subjects were female. There were 2063 patients with previous MI (41.53%), 225 with previous heart failure (HF) (13.39%), 679 with hypertension (23.77%), 265 with diabetes mellitus (10.36%), 202 with previous heart surgery (9.85%), either PTCA or CABG, and there were 1249 patients who were smokers.

Interventions. Nine beta blockers were investigated as treatment drugs compared with no intervention (three trials), placebo (three trials), CCBs (14), trimetazidine (two trials), and another kind of beta blocker (four trials). The duration of investigation ranged from 4 weeks to 84 months, with most trials lasting from 4 weeks to 3 months (18 trials). Only six trials lasted for longer than 1 year.

Outcomes. At least one interesting clinical outcome was demonstrated in these studies.

Publication bias and selecting reporting bias. It is unclear whether there was publication bias or selecting reporting bias. Only one trial explained its design, methodology, and management before the study began.⁸

Methodological quality of included studies. Details of the quality assessment are illustrated in Table 1. The study quality was generally assessed as a moderate. The quality of Chinese trials was considered as a high risk of bias. All studies were randomized. However, the appropriate randomization procedures were clearly described in only five of these studies.^{10,14,20,22,28} Three trials described the measures of allocation concealment.^{8,10,14} Nine trials demonstrated blinding methods adequately. No trials reported the loss of the follow up except for Harris et al.¹³ Withdrawal of the patients was reported in all these trials with a range from 3.70%³¹ to 25.04%.⁸ Five of these trials had performed intention-to-treat analysis.

Effects of interventions

Primary outcomes

All-cause mortality. Seven studies with total 2865 patients in the trials reported on all-cause mortality. Most were high-quality trials. Target patients were ≥ 62 years of age with New York Heart Association functional class 2 or 3, prior Q-wave MI, and an ejection fraction $\leq 40\%$. The beta blockers were found to

Table 1. Characteristics and methodological quality of included studies

Study	Population (T/C)	Treatment intervention	Control intervention	Duration	Outcome measures	Randomization	Allocation concealment	Blind	Loss to follow-up	Intention-to-treat analysis
Ahuja et al. (1993) ¹⁵	74/72	Metoprolol	Diltiazem	4 W	Ischaemic episode, side	U	U	U	Y	N
Ardissino et al. (1995) ¹⁶	145/135	Metoprolol	Nifedipine	6 W	MI, angina	U	U	D*	Y	N
Aronow et al. (1997) ¹⁷	79/79	Propranolol	Placebo	12 M	Death, MI, side	U	U	U	Y	Y
Boberg et al. (1992) ¹⁸	59/114	Atenolol	Epanolol	12 M	Death, side	U	U	D*	N	N
Cocco et al. (1979) ¹⁹	20/22	Pindolol	Nifedipine	45 D	Death, AMI, side	U	N	S	Y	N
Dargie et al. (1996) ⁸	286/325	Atenolol	Nifedipine	24 M	Death, UA, CABG, PTCA, treatment failure	U	Y	D**	Y	Y
De Vries et al. (1996) ²⁰	66/62	Atenolol	Nifedipine	4 W	Death, AMI, angina, GTN, side	Y	U	D*	Y	Y
Detry et al. (1994) ²¹	78/71	Propranolol	Trimetazidine	3 M	MI, angina, GTN, side	U	U	D*	U	N
Harris et al. (1983) ¹³	9/10	Timolol	Placebo	4 W	Death, AMI, angina, GTN, side	U	U	D*	Y	N
Hauf-Zachariou et al. (1997) ²²	126/122	Carvedilol	Verapamil	12 W	MI, angina, GTN, side	Y	U	D**	Y	Y
Hughes et al. (1990) ²³	18/19	Atenolol	Nicorandil	6 W	Death, side	U	U	D*	N	N
Kardas et al. (2007) ²⁴	49/53	Betaxolol	Metoprolol	8 W	QOL, side	U	N	N	Y	N
Kawanishi et al. (1992) ²⁵	38/36	Propranolol	Nifedipine	3 M	Angina, GTN	U	U	D**	Y	N
Mildtbo and Mølstad (2000) ²⁶	65/62	Metoprolol	Amlodipine	8 W	Death, AMI, angina, GTN, side	U	U	D**	Y	N
Pehrsson et al. (2000) ²⁷	116/116	Atenolol	Amlodipine	10 W	Angina, GTN, side	U	U	D**	Y	N
Pepine et al. (1994) ²⁸	152/154	Atenolol	Placebo	12 M	Death, MI, UA, revascularization, ischemic episode	Y	U	D**	Y	Y
Ping et al. (2003) ²⁹	49/48	Carvedilol	Atenolol	4 W	Angina, GTN, side	U	U	U	U	N
Rehnqvist et al. (1996) ⁹	406/403	Metoprolol	Verapamil	6–75 M	Death, AMI, revascularization, psychosomatic symptom, life satisfaction, sleep disturbance, side	U	U	D**	Y	N
Taylor et al. (1982) ¹⁷	839/621	Oxprenolol	Placebo	84 M	Death, MI	Y	Y	D**	N	N
Von Arnim (1995) ¹⁰	161/169	Bisoprolol	Nifedipine	4 W	Angina, ischaemic episode	Y	Y	D*	N	N

(continued)

Table 1. Continued

Study	Population (T/C)	Treatment intervention	Control intervention	Duration	Outcome measures	Randomization	Allocation concealment	Blind	Loss to follow-up	Intention-to-treat analysis
Wang et al. (2008) ³⁰	39/39	Metoprolol	Trimetazidine	10 W	Angina, GTN	U	U	U	Y	N
Wheatley (1985) ³¹	43/38	Atenolol	Diltiazem	6 W	Angina, GTN (2 W), side	U	U	D**	Y	N
Zhang et al. (2002) ³²	80/76	Carvedilol	Betaxolol	6 W	AMI, UA, side	U	U	U	Y	N
Zhang et al. (2006) ³³	36/35	Carvedilol	No	6 M	Angina	U	U	U	Y	N
Zhang et al. (2007) ³⁴	30/30	Metoprolol	Verapamil	8 W	Angina	U	U	U	N	N
Zhao et al. (2006) ³⁵	68/66	Betaxolol	No	6 M	UA	U	U	U	Y	N

AMI, acute myocardial infarction; C, control; CABG, coronary artery bypass grafting; D**, double blind and prescribed blind strategy; D*, double blind; D, day; GTN, glyceryl trinitrate consumption; M, month; N, no; PTCA, percutaneous transluminal coronary angioplasty; S, single blind; T, treatment; U, unsure; UA, unstable angina; W, week; Y, yes.

reduce the incidence of all-cause mortality significantly when compared with no treatment (OR 0.40, 95% CI 0.20 to 0.79).¹⁷ No significant difference was seen when the beta blockers were compared with placebo (OR 0.92, 95% CI 0.62 to 1.38).¹⁴ CCBs (OR 0.84, 95% CI 0.49 to 1.44; Table 2),^{8,9,26} nicorandil (a potassium channel opener; OR 0.30, 95% CI 0.01 to 7.83),²³ or another beta blocker (OR 0.97, 95% CI 0.09 to 10.87).¹⁸

Fatal and non-fatal AMI. Twelve trials with total 4044 patients in the studies reported the incidents of AMI. There was no significant reduction in the attack of AMI after treated with propranolol (OR 0.58, 95% CI 0.13 to 2.53).¹⁷ The risk of fatal and non-fatal AMI was not reduced when compared with placebo (Table 3), CCBs (Table 4), trimetazidine (OR 0.91, 95% CI 0.06 to 14.81),²¹ or between two beta blockers.^{18,32}

Secondary outcomes

Coronary revascularization. There was no significant reduction in the rate of revascularization when beta blockers were compared with placebo (OR 3.06, 95% CI 0.12 to 75.69)²⁸ or CCBs (overall OR 1.36, 95% CI 0.92 to 2.01)^{8,9} (Table 5).

Unstable angina. The higher incidence of UA was found in patients treated with atenolol compared placebo (OR 3.32, 95% CI 1.50 to 7.36).²⁸ However, the lower incidence of UA was seen in the patients treated with other beta blockers when compared with no control (overall OR 0.14, 95% CI 0.07 to 0.29).^{33,35} There was no significant reduction of UA when compared beta blockers with CCBs (Table 6), trimetazidine (OR 0.67, 95% CI 0.14 to 3.10),²¹ or between two beta blockers.^{18,32}

Angina attacks per week. No significant reduction in incidents of angina attacks per week was demonstrated when beta blockers were compared with placebo (mean difference, MD, -4.30, 95% CI -12.03 to 3.43),¹³ or with CCBs (Table 7), or with trimetazidine (WMD -0.62, 95% CI -2.28 to 1.04),^{21,30} or carvedilol was compared with atenolol (MD -0.41, 95% CI -0.75 to -0.07).²⁹

Consumption of glyceryl trinitrate per week. Most trials reported the consumption of GTN for patients with angina attacks. Two studies of these trials did not reported to treat the patients with angina attacks using nitrate consumption.^{10,34}

The significant decrease of the consumption of glyceryl trinitrate (GTN) per week was revealed when beta blockers were compared with CCBs (overall WMD -1.18, 95% CI -1.54 to -0.82)^{25,26} and carvedilol was compared

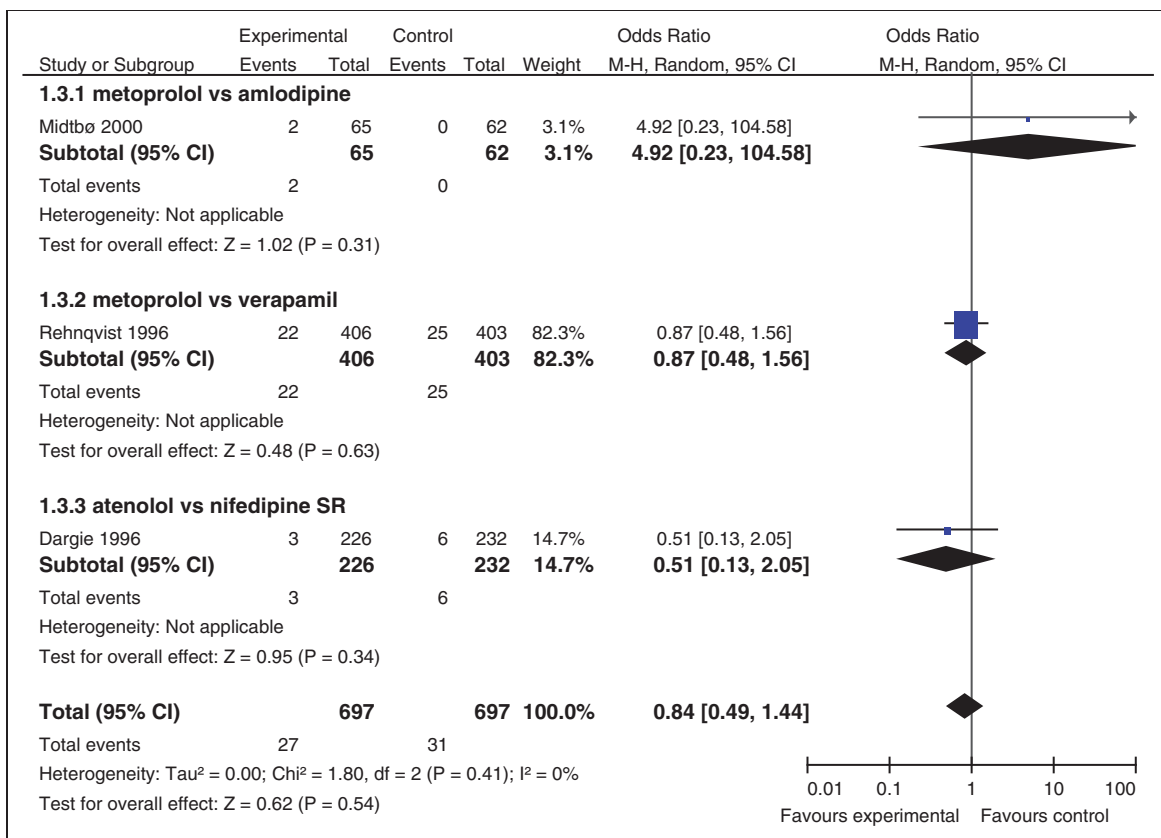
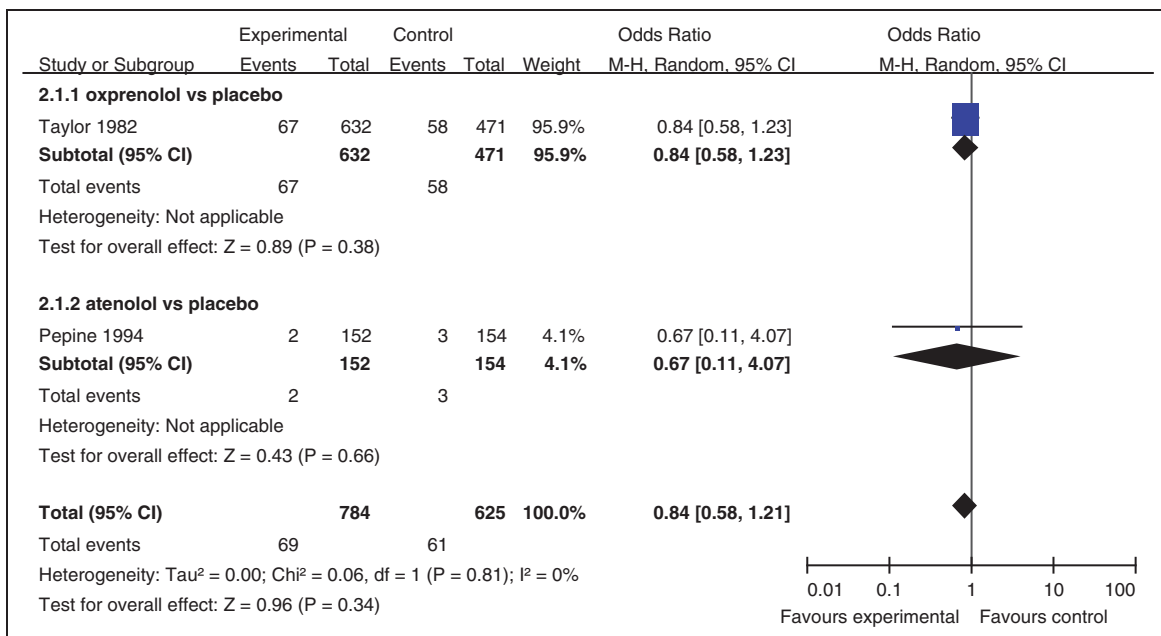
Table 2. All-cause mortality when compared with calcium-channel blockers**Table 3.** Acute myocardial infarction when compared with placebo

Table 4. Acute myocardial infarction when compared with calcium-channel blockers

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Odds Ratio M-H, Random, 95% CI	Odds Ratio M-H, Random, 95% CI
2.3.1 carvedilol vs verapamil							
Hauf-Zachariou 1997	0	125	2	122	1.9%	0.19 [0.01, 4.04]	
Subtotal (95% CI)		125		122	1.9%	0.19 [0.01, 4.04]	
Total events	0		2				
Heterogeneity: Not applicable							
Test for overall effect: Z = 1.06 (P = 0.29)							
2.3.2 metoprolol vs amlodipine							
Midtbø 2000	2	65	1	62	3.1%	1.94 [0.17, 21.91]	
Subtotal (95% CI)		65		62	3.1%	1.94 [0.17, 21.91]	
Total events	2		1				
Heterogeneity: Not applicable							
Test for overall effect: Z = 0.53 (P = 0.59)							
2.3.3 atenolol vs nifedipine SR							
Dargie 1996	14	226	15	232	31.9%	0.96 [0.45, 2.03]	
Subtotal (95% CI)		226		232	31.9%	0.96 [0.45, 2.03]	
Total events	14		15				
Heterogeneity: Not applicable							
Test for overall effect: Z = 0.12 (P = 0.91)							
2.3.4 metoprolol vs nifedipine							
Ardissino 1995	1	145	1	135	2.3%	0.93 [0.06, 15.03]	
Subtotal (95% CI)		145		135	2.3%	0.93 [0.06, 15.03]	
Total events	1		1				
Heterogeneity: Not applicable							
Test for overall effect: Z = 0.05 (P = 0.96)							
2.3.5 atenolol vs diltiazem							
Wheatley 1985	1	40	0	38	1.7%	2.92 [0.12, 74.01]	
Subtotal (95% CI)		40		38	1.7%	2.92 [0.12, 74.01]	
Total events	1		0				
Heterogeneity: Not applicable							
Test for overall effect: Z = 0.65 (P = 0.52)							
2.3.6 metoprolol vs verapamil							
Rehngvist 1996	29	406	25	403	59.0%	1.16 [0.67, 2.02]	
Subtotal (95% CI)		406		403	59.0%	1.16 [0.67, 2.02]	
Total events	29		25				
Heterogeneity: Not applicable							
Test for overall effect: Z = 0.53 (P = 0.59)							
Total (95% CI)		1007		992	100.0%	1.08 [0.71, 1.66]	
Total events	47		44				
Heterogeneity: Tau ² = 0.00; Chi ² = 2.01, df = 5 (P = 0.85); I ² = 0%							
Test for overall effect: Z = 0.37 (P = 0.71)							

0.01 0.1 1 10 100

Favours experimental Favours control

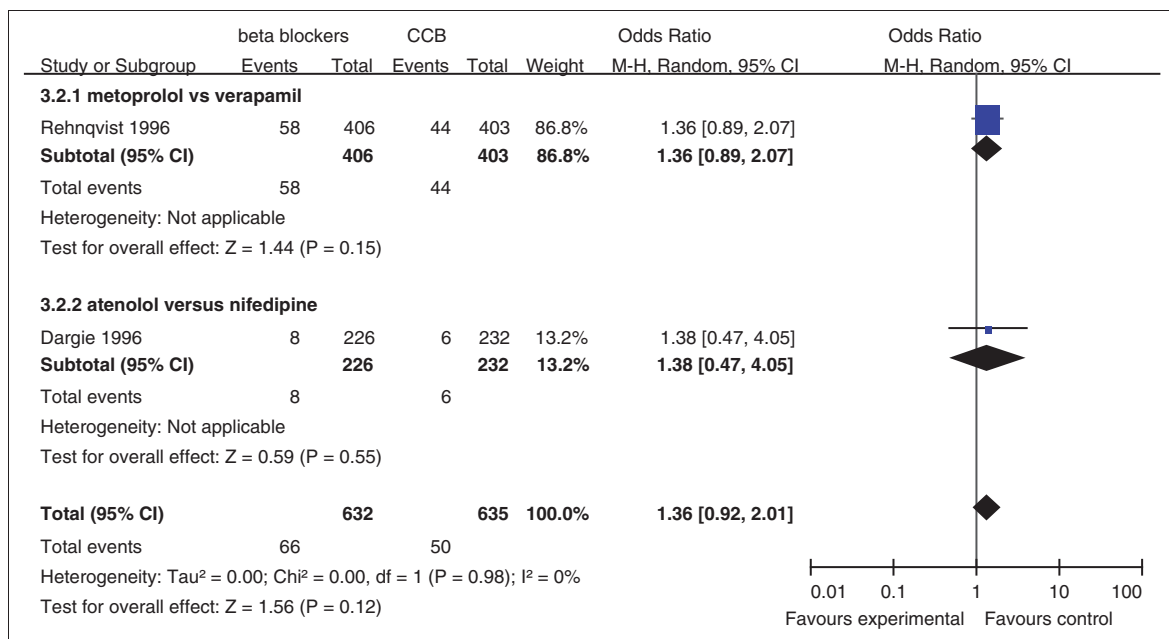
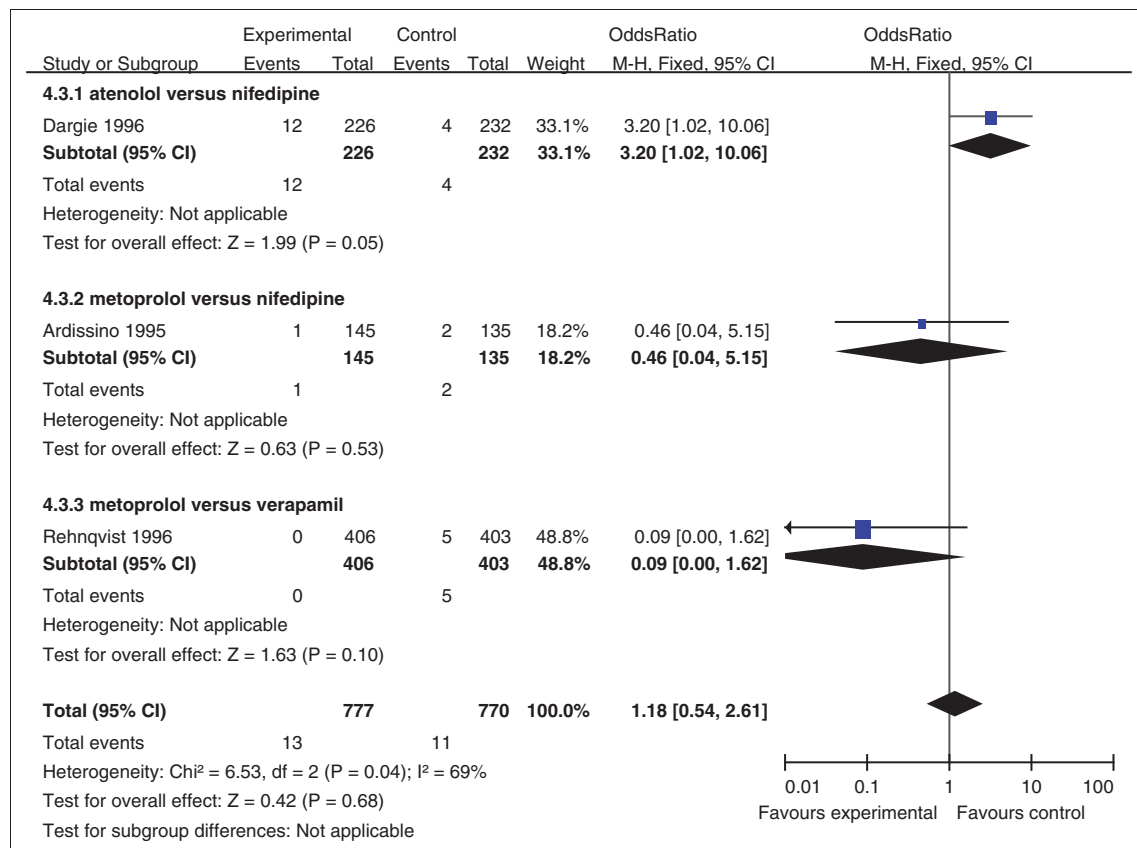
Table 5. Coronary revascularization when compared with calcium-channel blockers**Table 6.** Unstable angina when compared with calcium-channel blockers

Table 7. Angina attacks per week when compared with calcium-channel blockers

Study or Subgroup	Experimental			Control			Mean Difference		Mean Difference
	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI	IV, Random, 95% CI
5.2.1 propranolol vs nifedipine (parallel)									
Kawanishi 1992	3.2	6.1	38	4.3	6.4	36	10.2%	-1.10 [-3.95, 1.75]	
Subtotal (95% CI)			38			36	10.2%	-1.10 [-3.95, 1.75]	
Heterogeneity: Not applicable									
Test for overall effect: Z = 0.76 (P = 0.45)									
5.2.2 metoprolol vs amlodipine (parallel)									
Midtbø 2000	3.1	1.1	60	4.7	1.2	57	34.3%	-1.60 [-2.02, -1.18]	
Subtotal (95% CI)			60			57	34.3%	-1.60 [-2.02, -1.18]	
Heterogeneity: Not applicable									
Test for overall effect: Z = 7.51 (P < 0.00001)									
5.2.3 metoprolol vs verapamil (parallel)									
Zhang 2007	-2.54	0.69	25	-2.38	0.75	25	34.4%	-0.16 [-0.56, 0.24]	
Subtotal (95% CI)			25			25	34.4%	-0.16 [-0.56, 0.24]	
Heterogeneity: Not applicable									
Test for overall effect: Z = 0.78 (P = 0.43)									
5.2.4 bisoprolol vs nifedipine (parallel)									
von Arnim 1995	2.8	5.4	133	4.4	7.1	135	21.1%	-1.60 [-3.11, -0.09]	
Subtotal (95% CI)			133			135	21.1%	-1.60 [-3.11, -0.09]	
Heterogeneity: Not applicable									
Test for overall effect: Z = 2.08 (P = 0.04)									
Total (95% CI)			256			253	100.0%	-1.05 [-2.13, 0.02]	
Heterogeneity: Tau ² = 0.83; Chi ² = 24.79, df = 3 (P < 0.00001); I ² = 88%									
Test for overall effect: Z = 1.92 (P = 0.05)									

Favours experimental Favours control

Favours experimental Favours control

with atenolol (MD -0.72 , 95% CI -1.34 to -0.10).²⁹ No difference of the GTN consumption was demonstrated between beta blocks and placebo (MD -4.20 , 95% CI -13.15 to 4.75),¹³ or between beta blocks and trimetazidine (WMD -0.32 , 95% CI -0.69 to 0.05).^{21,30}

Quality of life. The psychosomatic score (inter-group: $p = 0.34$) and the estimate of sleep disturbances ($p = 0.97$) were improved similarly in both metoprolol and verapamil groups, whereas overall QOL were not changed during follow-up ($p = 0.85$).⁹ The total follow-up period was 2887 patient years (median follow-up time 3.4 years).

Midtbø and Mølsted found that improvements in general well-being and physical performance were demonstrated in the patients' groups treated with either metoprolol or amlodipine, with no significant difference between these two groups ($p = 0.43$).²⁶ There was also no difference in general health-related QOL between betaxolol and metoprolol.²⁴

Side effects. The number of patients who suffered adverse side effects was reported in 13 of these trials.

The clinical manifestation of those side effects varied from each study. In general for the beta blocker groups, side effects were more common in tiredness, congestive HF, and cardiac arrest.

Side effects were mild and similar both in beta blocker groups and placebo (overall OR 0.75, 95% CI 0.08 to 6.93),^{13,28} but more common in propranolol group than in no beta blocker group (OR 26.69, 95% CI 1.54 to 461.38),¹⁷ and no difference when compared with CCBs (overall OR 0.92, 95% CI 0.71 to 1.19),^{9,20,22,26,27,31} with trimetazidine,²¹ or with nicorandil,²³ or between two beta blockers.^{24,29,32}

Discussion

We have obtained the clinical information of 6108 patients with stable angina pectoris from total 26 trials. These patients with stable angina pectoris presented with different stages or severity of the disease or accompanied with variety of other medical conditions. Based on the data obtained from these studies, we have investigated the effectiveness and safety using

beta blockers as treatment for patients with stable angina pectoris. Of the total 26 studies, study quality was assessed as moderate. This could potentially have biased the results in favour of treatment.

Our systematic review has found that there are no reductions of mortality, AMI and angina attacks, no improvement of QOL, and no more side effects when comparing beta blockers with any other anti-anginal drugs or placebo. When beta blockers were compared with no control, significant reductions in mortality and UA with less side effects have been demonstrated. There is less GTN consumption in patients treated with beta blockers than in those treated with CCBs.

Globally, results for included outcomes were unsatisfactory and susceptible to bias due to the fact that most of them were extracted from only several studies. Some studies had small sample sizes and short duration of follow-up and were of lower quality. Thus, further large, long-term studies are required to draw conclusions. Some trials did not assess the most important outcomes defined in this review. Because most crossover design trials had very short duration of observation, we cannot draw any conclusion on mortality, AMI, and UA. The information of crossover trials is not included in our analysis.

We have measured the degree of inconsistency among these studies. Heterogeneity in response did not appear to result from differences in methods used for the primary outcomes. Some of the secondary outcomes were settled based on the different interventions and different measuring unit of studies. The heterogeneity could be caused by differences in samples, methods, results expression, duration of observation, and medications.

In our systematic review, we have found no benefit on mortality. Our finding is different from that of Freemantle et al., who identified a significant reduction in death in beta blockers compared with control, especially in long-term trials.³⁶ In another report, the 25% reduction in mortality has been observed in patients using beta blockers as long-term treatment during and after MI.⁶ These different findings may be caused by different methods for recruiting patients. We included not only patients with stable angina, but also those after MI. We think that a firm conclusion on all-cause mortality cannot be made because of limited evidence.

One study has showed that the incidences of mortality were significantly reduced in patients with chronic HF and prior MI using beta blockers than in those using no control.¹⁷ This finding may indicate that the patients with previous MI using beta blockers as treatment could give more benefits.

Beta blockers are useful agents in the prevention and relief of angina pain. In our systematic review, we have demonstrated that beta blockers do not decrease angina attacks significantly. This result is similar to that of Yusuf et al.⁶

Our data indicate that there were no differences in QOL clinically when beta blockers were compared with CCBs. Kardas et al.²⁴ has proved that once-daily dosing of beta blockers leads to significantly better compliance than twice-daily doses. Similar results have been also observed when beta blockers are used in patients with hypertension.

In our systematic review, there are no significant differences between different beta blockers, either for clinical benefits or for adverse events. Yusuf et al. also did not find important differences clearly apparent among the benefits of different beta blockers.⁶

Adverse effects are as important as the efficacy of the interventions in clinical study. This systematic review indicates that beta blockers are well tolerated. No adverse events are occurred in patients treated with beta blockers when compared with patients using placebo or other anti-anginal agents. This result is similar with the findings of another systematic investigation, which identified that beta blockers provide fewer adverse events than CCBs.¹¹ It has been shown that treatment with beta blockers can be maintained over long periods of time.¹⁴

None of the trials in our review reported the economic costs in the treatment of disease. In future, more studies using beta blockers as a long-term treatment programme for stable angina pectoris should be done to evaluate the cost-effectiveness ratio.

Conclusions

Implications for practice

Our data suggest that beta blockers may have an incremental effect on prophylaxis of myocardial ischaemia in stable angina patients compared with no treatment, but no more effective than other anti-anginal agents. However, the evidence is weak and caution is required in interpreting this result because most of the outcomes were extracted from only several studies.

Implications for research

It is disappointing that a comprehensive study in the outcomes and side effects over longer duration of treatment with beta blockers has not been reported. Further trials should aim to follow patients up for at least 5 years and report more clinical outcomes. Outcome index units should be identical.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

References

1. Braunwald E. *Heart disease: A textbook of cardiovascular medicine*, 4th edn. Philadelphia/London: Saunders, Douglas P Zipes, 1992.
2. Nielsen KM, Foldspang A, Larsen ML, Gerdes LU, Rasmussen S and Faergeman O. Estimating the incidence of the acute coronary syndrome: data from a Danish cohort of 138 290 persons. *Eur J Cardiovasc Prev Rehabil* 2007; 14(5): 608–614.
3. American Heart Association. *Heart disease and stroke statistics: 2004 update*. Dallas, TX: American Heart Association, 2003.
4. Agency for Health Care Policy and Research. AHCPR guidelines revised by KAU. Unstable angina: diagnosis and management. <http://www.medana.unibas.ch/eng/internt/angina.htm> (1996, accessed 22 December 2003).
5. The American College of Cardiology and the American Heart Association Task Force on Practice Guidelines (Committee on Management of Patient With Chronic Stable Angina). Guidelines for the management of patients with chronic stable angina. *J Am Coll Cardiol* 1999; 33(7): 2092–2197.
6. Yusuf S, Peto R, Lewis J, Collins R and Sleight P. Beta blockade during and after myocardial infarction: an overview of the randomized trials. *Prog Cardio Dis* 1985; 27(5): 335–371.
7. Reiner Z, Mihatov S, Milicic D, Bergovec M and Planinc D. TASPIC-CRO Study Group Investigators. Treatment and secondary prevention of ischemic coronary events in Croatia (TASPIC-CRO study). *Eur J Cardiovasc Prev Rehabil* 2006; 13(4): 646–654.
8. Dargie HJ, Ford I and Fox KM. Total Ischaemic Burden European Trial (TIBET). Effects of ischaemia and treatment with atenolol, nifedipine SR and their combination on outcome in patients with chronic stable angina. The TIBET Study Group. *Eur Heart J* 1996; 17(1): 104–112.
9. Rehnqvist N, Hjendahl P, Billing E, Bjorkander I, Eriksson SV, Forslund L, et al. Effects of metoprolol vs verapamil in patients with stable angina pectoris. The Angina Prognosis Study in Stockholm (APSIS). *Eur Heart J* 1996; 17(1): 76–81.
10. von Arnim T. Medical treatment to reduce total ischemic burden: total ischemic burden bisoprolol study (TIBBS), a multicenter trial comparing bisoprolol and nifedipine. The TIBBS investigators. *J Am Coll Cardiol* 1995; 25(1231): 231–238.
11. Heidenreich PA, McDonald KM, Hastie T, Fadel B, Hagan V, Lee BK, et al. Meta-analysis of trials comparing beta-blockers, calcium antagonists, and nitrates for stable angina. *JAMA* 1999; 281(20): 1927–1936.
12. Higgins JPT and Green S. *Cochrane handbook for systematic reviews of interventions*, version 5.0.0 edition. Oxford: The Cochrane Collaboration, 2008.
13. Harris FJ, Low RI, Paumer L, Amsterdam EA and Mason DT. Antianginal efficacy and improved exercise performance with timolol. Twice-daily beta blocker in ischemic heart disease. *Am J Cardiol* 1983; 51(1): 13–18.
14. Taylor SH, Silke B, Ebbutt A, Sutton GC, Prout BJ and Burley DM. A long-term prevention study with oxprenolol in coronary heart disease. *N Engl J Med* 1982; 307(21): 1293–1301.
15. Ahuja RC, Sinha N, Kumar RR and Saran RK. Effect of metoprolol and diltiazem on the total ischaemic burden in patients with chronic stable angina: a randomised controlled trial. *Int J Cardiol* 1993; 41(3): 191–199.
16. Ardissino D, Savonitto S, Egstrup K, Rasmussen K, Bae EA, Omland T, et al. Selection of medical treatment in stable angina pectoris: results of the International Multicenter Angina Exercise (IMAGE) Study. *J Am Coll Cardiol* 1995; 25(7): 1516–1521.
17. Aronow WS, Ahn C and Kronzon I. Effect of propranolol versus no propranolol on total mortality plus non-fatal myocardial infarction in older patients with prior myocardial infarction, congestive heart failure, and left ventricular ejection fraction less than or equal to 40% treated with diuretics plus angiotensin-converting enzyme inhibitors. *Am J Cardiol* 1997; 80(2): 207–209.
18. Boberg J, Larsen FF and Pehrsson SK. The effects of beta blockade with (epanolol) and without (atenolol) intrinsic sympathomimetic activity in stable angina pectoris. The Visacor Study Group. *Clin Cardiol* 1992; 15(8): 591–595.
19. Cocco G, Strozzi C, Chu D, Amrein R and Castagnoli E. Therapeutic effects of pindolol and nifedipine in patients with stable angina pectoris and asymptomatic resting ischemia. *Eur J Cardiol* 1979; 10(1): 59–69.
20. de Vries RJ, van den Heuvel AF, Lok DJ, Claessens RJ, Bernink PJ, Pasteuning WH, et al. Nifedipine gastrointestinal therapeutic system versus atenolol in stable angina pectoris. *Int J Cardiol* 1996; 57(2): 143–150.
21. Detry JM, Sellier P, Pennaforte S, Cokkinos D, Dargie H and Mathes P. Trimetazidine: a new concept in the treatment of angina. Comparison with propranolol in patients with stable angina. Trimetazidine European Multicenter Study Group. *Br J Clin Pharmacol* 1994; 37(3): 279–288.
22. Hauf-Zachariou U, Blackwood RA, Gunawardena KA, O'Donnell JG, Garnham S and Pfarr E. Carvedilol versus verapamil in chronic stable angina: a multicentre trial. *Eur J Clin Pharmacol* 1997; 52(2): 95–100.
23. Hughes LO, Rose EL, Lahiri A and Raftery EB. Comparison of nicorandil and atenolol in stable angina pectoris. *Am J Cardiol* 1990; 66(7): 679–682.
24. Kardas P. Compliance, clinical outcome, and quality of life of patients with stable angina pectoris receiving once-daily betaxolol versus twice daily metoprolol: a randomized controlled trial. *Vasc Health Risk Manag* 2007; 3(2): 235–242.
25. Kawanishi DT, Reid CL, Morrison EC and Rahimtoola SH. Response of angina and ischemia to long-term treatment in patients with chronic stable angina: a double-blind randomized individualized dosing trial of nifedipine, propranolol and their combination. *J Am Coll Cardiol* 1992; 19(2): 409–417.
26. Midtbø K and Mølsted P. Amlodipine versus slow release metoprolol in the treatment of stable exertional

- angina pectoris (AMSA). *Scand Cardiovasc J* 2000; 34(5): 475–479.
27. Pehrsson SK, Ringqvist I, Ekdahl S, Karlson BW, Ulvenstam G and Persson S. Monotherapy with amlodipine or atenolol versus their combination in stable angina pectoris. *Clin Cardiol* 2000; 23(10): 763–770.
 28. Pepine CJ, Cohn PF, Deedwania PC, Gibson RS, Handberg E, Hill JA, et al. for the ASIST Study Group. Effects of treatment on outcome in mildly symptomatic patients with ischemia during daily life. the Atenolol Silent Ischemia Study (ASIST). *Circulation* 1994; 90(2): 762–768.
 29. Ping YH, Shen LT, Hu FH and Liu HL. Efficacy of carvedilol in the treatment of stable exertional angina pectoris. *Chin J Cardiovasc Med* 2003; 8(1): 29–31.
 30. Wang HY, Wang LL and Wang S. Effect of trimetazidine and metoprolol on the stable angina pectoris patients with type 2 diabetes mellitus. *Clin J Prim Med Pharm* 2008; 15(6): 889–890.
 31. Wheatley D. A comparison of diltiazem and atenolol in angina. *Postgrad Med J* 1985; 61(719): 785–789.
 32. Zhang LH, Xue YS, Zhao JH, Liu J, Zhen QS, Xiao JP, et al. Efficacy of carvedilol compared with metoprolol in stable angina. *Chin Heart J* 2002; 14(6): 488–490.
 33. Zhang 2006 Zhang BL and Chang LX. Clinical observation of carvedilol for stable angina. *Pract J Cardiac Cerebr Pnuemon Vasc Dis* 2006; 14(9): 729.
 34. Zhang Q, Lu XN and Sun NL. Effects of verapamil and metoprolol on heart rate variability in patients with coronary heart disease. *Beijing Da Xue Xue Bao* 2007; 39(6): 610–613.
 35. Zhao ZX and Song CY. Clinical analysis of β blockers for 68 patients with stable angina. *J Chin Modern Med* 2006; 3(7): 805.
 36. Freemantle N, Cleland J, Young P, Masson J and Harrison J. Beta blockade after myocardial infarction: systematic review and meta regression analysis. *BMJ* 1999; 318(7200): 1730–1737.